Solution

λ = 5000 Å = 5000×10^{-10} m; sin 30° = 0.5; n = 1; a =?

Equation for diffraction minimum is, $a\sin\theta = n\lambda$

The central maximum is spread up to the first minimum. Hence, n = 1

Rewriting, $a = \dfrac{\lambda}{\sin\theta}$

Substituting, $a = \dfrac{5000\times10^{-10}}{0.5}$

$a = 1\times10^{-6}\,\text{m} = 0.001\times10^{-3}\,\text{m} = 0.001\,\text{mm}$

7.4.3 Discussion on first minimum

The equation for first minimum in single slit diffraction is, a sin θ = λ. The angular spread for its first minimum in the diffraction pattern is, $\sin\theta = \dfrac{\lambda}{a}$. The central maximum is found in between these first minima that occur on both the sides. We can discuss the following cases on the central maximum.

(i) If $a < \lambda$, then $\sin\theta > 1$ which is not possible. Hence, diffraction does not take place.

(ii) If $a = \lambda$, then sin θ = 1 i.e. θ = 90°. The first minimum is at 90°. Hence, the central maximum spreads fully into the geometrically shadowed region leading to the bending of the diffracted light by 90°.

(iii) If $a > \lambda$ and also comparable to λ, say $a = 2\lambda$, then $\sin\theta = \dfrac{1}{2}$ (or) θ = 30°. The diffraction is observed with a measurable spread. Hence, it is concluded that for observing the diffraction pattern, essentially the width of the slit a must be just few times greater than the wavelength of light λ.

(iv) If $a >> \lambda$, then sin $\theta << 1$ i.e. The first minimum falls within the width space of the slit itself. Hence, the phenomenon of diffraction is not observed at all.

7.4.4 Fresnel's distance

The rectilinear propagation of light is violated as there is bending of light in diffraction. But, this bending is not seen till the diffracted ray crosses the central maximum at a distance z from the slit as shown in Figure 7.19. Hence, **Fresnel's distance is the distance upto which the ray optics is obeyed and beyond which the ray optics is not obeyed; but, the wave optics becomes significant.**

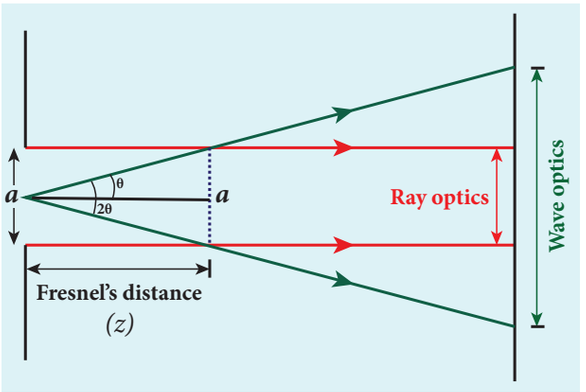


Figure 7.19 Fresnel's distance

The diffraction equation for first minimum is, $\sin\theta = \dfrac{\lambda}{a}$; when θ is small, $\theta = \dfrac{\lambda}{a}$

From the definition of Fresnel's distance, $2\theta = \dfrac{a}{z}$ (or) $\theta = \dfrac{a}{2z}$

Equating the above two equation for θ gives , $\dfrac{\lambda}{a} = \dfrac{a}{2z}$

After rearranging, we get Fresnel's distance z as,

$$z = \frac{a^2}{2\lambda} \qquad (7.45)$$

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